Limitations in intense exercise performance of athletes – Effect of speed endurance training on ion handling and fatigue development

Article *in* The Journal of Physiology · September 2016

DOI: 10.1113/JP273218

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The Journal of Physiology

TOPICAL REVIEW

Limitations in intense exercise performance of athletes – effect of speed endurance training on ion handling and fatigue development

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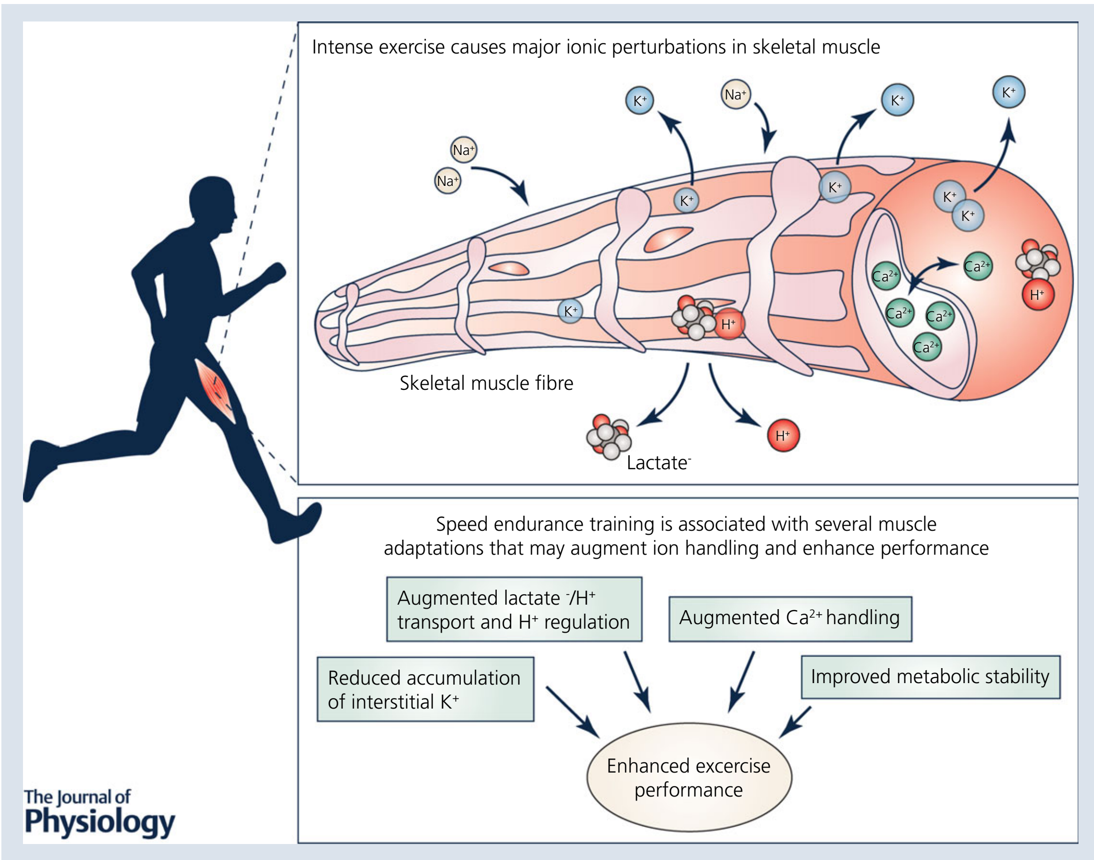

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 DOI: 10.1113/JP273218

Abstract Mechanisms underlying fatigue development and limitations for performance during intense exercise have been intensively studied during the past couple of decades. Fatigue development may involve several interacting factors and depends on type of exercise undertaken and training level of the individual. Intense exercise (½–6 min) causes major ionic perturbations (Ca^{2+}, Cl^-, H^+, K^+, lactate$^-$ and Na^+) that may reduce sarcolemmal excitability, Ca^{2+} release and force production of skeletal muscle. Maintenance of ion homeostasis is thus essential to sustain force production and power output during intense exercise. Regular speed endurance training (SET), i.e. exercise performed at intensities above that corresponding to maximum oxygen consumption ($\dot{V}_{O_2,max}$), enhances intense exercise performance. However, most of the studies that have provided mechanistic insight into the beneficial effects of SET have been conducted in untrained and recreationally active individuals, making extrapolation towards athletes' performance difficult. Nevertheless, recent studies indicate that only a few weeks of SET enhances intense exercise performance in highly trained individuals. In these studies, the enhanced performance was not associated with changes in $\dot{V}_{O_2,max}$ and muscle oxidative capacity, but rather with adaptations in muscle ion handling, including lowered interstitial concentrations of K^+ during and in recovery from intense exercise, improved lactate$^-$–H^+ transport and H^+ regulation, and enhanced Ca^{2+} release function. The purpose of this Topical Review is to provide an overview of the effect of SET and to discuss potential mechanisms underlying enhancements in performance induced by SET in already well-trained individuals with special emphasis on ion handling in skeletal muscle.

(Received 29 July 2016; accepted after revision 21 September 2016; first published online 27 September 2016)
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Abstract figure legend Intense exercise causes major ionic perturbations (Ca^{2+}, Cl^-, H^+, K^+, lactate$^-$ and Na^+) that may reduce sarcolemmal excitability, Ca^{2+} release, and force production of skeletal muscle. Maintenance of ion homeostasis is thus essential to sustain force production and power output during intense exercise. Speed endurance training, in which training sessions are performed at intensities above that corresponding to maximum oxygen consumption, is an efficient way to enhance intense exercise performance in already well-trained individuals. The enhancements in performance during intense exercise induced by speed endurance training may be related to muscle adaptations in ion handling, including improved regulation of K^+ homeostasis, enhanced lactate$^-$–H^+ transport, and in some instances augmented H^+ buffering capacity, sarcoplasmic reticulum Ca^{2+} handling and glycolytic ATP provision.

Abbreviations ClC-1, Cl^- channel isoform 1; FXYD1, phospholemman; K_{IR}2.1, strong inward rectifying K^+ channel; K_{ATP}, ATP-sensitive K^+ channel; K_{IR}6.2, inward rectifying channel K^+ subunit 6.2; MCT1, monocarboxylate cotransporter isoform 1; MCT4, monocarboxylate cotransporter isoform 4; NHE1, Na^+–H^+ exchanger isoform 1; NKCC1, Na^+–K^+–$2Cl^-$ exchanger isoform 1; ROS, reactive oxygen species; RyR1, ryanodine receptor isoform 1; SERCAI, sarcoplasmic reticulum Ca^{2+} ATPase isoform 1; SERCAII, sarcoplasmic reticulum Ca^{2+} ATPase isoform 2; SET, speed endurance training; SR, sarcoplasmic reticulum; $\dot{V}_{O_2,max}$, maximum oxygen consumption.

Introduction

Training is fundamental to improve exercise performance in the athlete. Optimization of training strategies for performance-enhancements is thus an important area of research within exercise physiology and sports medicine (Faria *et al.* 2005; Laursen, 2010; Bishop *et al.* 2011). In recent years, much research has focused on speed endurance training (SET), with brief intervals ($\leq$ 60 s) performed at intensities above that corresponding to maximum oxygen consumption ($\dot{V}_{O_2,max}$) (Ross & Leveritt, 2001; Iaia & Bangsbo, 2010; Gibala & Jones, 2013; Bangsbo, 2015). As presented in Table 1, SET is an efficient way to enhance intense exercise performance (½–6 min) in already well-trained individuals and athletes, even with marked reductions in weekly training volume. It should be emphasised that in several of the studies presented in Table 1, training was also intensified by introducing or extending aerobic high-intensity training, which is characterized by interval training with intensities close to those eliciting $\dot{V}_{O_2,max}$.

The mechanisms underlying performance enhancements induced by a period of SET are not completely clear. Although several advanced human physiological experiments have been undertaken to investigate the effects of SET, a limitation of the research within this area is that most studies included untrained or recreationally active individuals (Iaia & Bangsbo, 2010; Gibala &

Table 1. Adaptations to SET in performance during intense exercise (½–6 min) and skeletal muscle of trained individuals

Study	Subjects (no., M/F, exercise type)	$\dot{V}_{O_2,max}$	Intervention (duration, sets/reps, work/rest, frequency, Δ% volume)	Performance measure	Effect (Δ%)	Muscle adaptation	Effect (Δ%)
Houston & Thomson, 1977	5 M runners	59	6 wks of 3×60 s/120 s + 5×6 s/24 s + 2×90 s/180 s (4/wk) + leg press (15 RM)	Intense exercise (∼1 min) 60 s running distance 90 s running distance	+17 +14 +13	ATP content	+15
Nevill *et al.* 1989	4 M/4F runners	55	8 wks of 2×30 s/10 min + 6–10×6 s/54 s + 2–5×2 min/5 min (3–4/wk)	30 s all out	+6	Rate of anaerobic glycolytic ATP production *In vivo* buffering capacity	+14 +44
Shepley *et al.* 1992	9 M runners	67	1 wk of 1–5×∼70 s/6–12 min (5/wk) −91% training volume	Intense exercise (∼5 min)	+22	CS	+18
Dawson *et al.* 1998	9 M trained	57	6 wks of 3–5×4-8×10 s/40–60 s (3/wk)	Repeated sprint Intense exercise (∼1 min)	+2 +11	Phosphorylase activity CS activity	+40 −36
Ørtenblad *et al.* 2000	9 M trained	61	5 wks of 20×10 s/50 s (3/wk)	Repeated sprint	+12	Ca^{2+} release SERCAI SERCAII RyR1	+9 +41 +55 +48
Bickham *et al.* 2006	7 M runners	58	6 wks of 4 × 14–30 × 40–100 m (3/wk)	Intense exercise (∼2½ min)	+11	Type II fibres MCT1	+6 +50
Bangsbo *et al.* 2009	12 M runners	63	6-9 wks.of 8–12×30 s/180 s (2–3/wk) + 4×4 min (1–2/wk) −30%, training volume	Intense exercise (∼2 min) 30 s all-out	+36 +5–7	Na^+–K^+ subunit α_2 Na^+–K^+ subunit β_1	+68 (+10)
Iaia *et al.* 2008, 2009	8/9 M runners	56	4 wks of 8–12×30 s/180 s (3–4/wk) −64% training volume	Intermittent running Intense exercise (∼1½ min) 30 s all-out	+19 +19–27 +7	Na^+–K^+ subunit α_1 Na^+–K^+ subunit α_2 NKCC1 NHE1	+29 (+16) (+14) +30
Thomassen *et al.* 2010; Christensen *et al.* 2011	7 M football	55	2 wks of 10–12×30 s/180 s (2/wk) −30% training volume	Repeated sprint Intermittent running	+2 (+6)	Na^+–K^+ subunit α_2 FXYD1-phos MCT1 PDH	+15 +27 (+13) +17
Gunnarsson *et al.* 2012	7 M football	60	5 wks of 5–9×30 s/180 s (1/wk) +11% training volume	Intermittent running	+11	MCT1 Na^+–K^+ subunit β_1	+9 −13

(*Continued*)

Table 1. continued

Study	Subjects (no., M/F, exercise type)	$\dot{V}_{O_2,max}$	Intervention (duration, sets/reps, work/rest, frequency, Δ% volume)	Performance measure	Effect (Δ%)	Muscle adaptation	Effect (Δ%)
Gunnarsson *et al.* 2013; Thomassen *et al.* 2016	8 M cyclists	59	7 wks of 10–12×30 s/270 s (2–3/wk)+ 4–5×3-4 min (1–2/wk) −70% training volume	Repeated sprint	+4	*In vivo* buffering capacity	+18
				Intense exercise (∼4 min)	+18	CS	−16
						COX4	−11
						K_{IR}2.1	(+18)
						K_{IR}6.2	−14
						NHE1	(+30)
						FXYD1	+30
						FXYD1-phos	+40
						PLN	+16
						PLN-phos	+42
						CaMKII	+25
						CaMKII-phos	+300
Puype *et al.* 2013	9 M trained	53	6 wks of 4–9×30 s/270 s (3/wk)	–		PFK	+17
						MCT1	+70
Skovgaard *et al.* 2014	12 M runners	59	8 wks of 4–12×30 s/180 s + resistance training (4/wk) −42% training volume	1500 m running (∼5 min)	+5	NHE1	+35
				Intermittent running	+44	SERCAI	−15
Nyberg *et al.* 2016	16 M football	–	9 wks of 2–3×8–10×5 s/10 s (1½/wk)	Intermittent running	+12	HAD	−8
						COX4	−9
						OXPHOS	−16
						Capillarization	−15
Vorup *et al.* 2016	9 M runners	60	8 wks of 4–8×30 s/180 s (2/wk) + resistance training/aerobic moderate or high intensity training (2/wk) −58% training volume	400 m running (∼1 min)	+5	Na^+–K^+ subunit β_1	+15
				Intermittent running	+19	LDH	+17

Relative effect size for performance and muscle changes from each study are only presented for statistical significant measures ($P \le 0.05$) or when trending towards significance ($P = 0.05$–0.10) in parentheses. Abbreviations: CaMKII, Ca^{2+}–calmodulin-dependent protein kinase II; COX4, cytochrome *c* oxidase subunit 4; CS, citrate synthase, HAD, hydroxyacyl-CoA dehydrogenase, K_{IR}2.1, K^+ inward rectifier channel 2.1; K_{IR}6.2, K^+ inward rectifier channel 6.2; LDH, lactate dehydrogenase; MCT, monocarboxylate transporter; NHE1, Na^+–H^+ exchanger isoform 1; NKCC1, Na^+–K^+–$2Cl^-$ channel isoform 1; OXPHOS, oxidative phosphorylation complex I–V enzymes; PFK, phosphofructokinase; PLN, phospholamban; RyR1, ryanodine receptor 1, SERCA, sarcoplasmic reticulum Ca^{2+} ATPase; $\dot{V}_{O_2,max}$, maximum oxygen consumption (ml kg^{-1} min^{-1}).

Jones, 2013; Weston *et al.* 2014; Bangsbo, 2015), which makes translation towards athletes' performance difficult. In untrained and recreationally active individuals, SET augments $\dot{V}_{O_2,max}$ by 7–15% after only few weeks of training despite a low volume of training (Burgomaster *et al.* 2008; Jacobs *et al.* 2013). This observation is not surprising, since the heart rate of the individuals during a SET session with 6–10 × 30 s sprints, separated by 90 s of recovery, may reach mean and peaks of around 85 and 95% of maximum heart rate, respectively (Mohr *et al.* 2007). Furthermore, faster $\dot{V}_{O_2}$ kinetics, increased capillarization, and augmented mitochondrial function of skeletal muscle have been observed in untrained and recreationally active individuals following a period of SET (Jensen *et al.* 2004; Gibala *et al.* 2006; Jacobs *et al.* 2013; Christensen *et al.* 2016). In contrast, SET does not increase $\dot{V}_{O_2,max}$, capillarization or oxidative capacity of skeletal muscle in already trained individuals (Bangsbo *et al.* 2009; Iaia *et al.* 2009; Christensen *et al.* 2015). Actually, SET has been shown to enhance exercise performance in spite of a concurrent reduction in expression of oxidative enzymes and capillarization of skeletal muscle in well-trained individuals (Dawson *et al.* 1998; Nyberg *et al.* 2016). On the other hand, performance enhancements in trained individuals elicited by a period of SET are associated with several adaptations related to ion handling in skeletal muscle. SET may augment K^+ handling (Bangsbo *et al.* 2009; Thomassen *et al.* 2010, 2016), lactate$^-$–H^+ transport capacity (Bickham *et al.* 2006; Gunnarsson *et al.* 2012, 2013; Puype *et al.* 2013), H^+ regulation (Nevill *et al.* 1989; Gunnarsson *et al.* 2013; Skovgaard *et al.* 2014), and Ca^{2+} handling function (Ørtenblad *et al.* 2000) of trained individuals. The importance of these adaptations for enhancements in performance during intense exercise (½–6 min) following a period of SET in already trained individuals will be discussed in the present Topical Review.

Development of fatigue during intense exercise

Skeletal muscle fatigue can be defined as a disruption of the force production needed to meet the demand for a given exercise intensity (McKenna *et al.* 2008). The mechanisms underlying muscle fatigue development are highly debated and complex (Allen *et al.* 2008; Cairns & Lindinger, 2008; McKenna *et al.* 2008; MacIntosh & Shahi, 2011). Thus, fatigue development may involve several interacting factors and also depends on exercise modality and inter-individual differences (Knicker *et al.* 2011). Intense exercise (½–6 min) causes major metabolic and ionic perturbations (Ca^{2+}, Cl^-, H^+, K^+, lactate$^-$, Mg^{2+}, Na^+ and P_i) that impair excitation–contraction coupling of skeletal muscle, decrease mechanical efficiency and lead to force decline (i.e. skeletal muscle fatigue) (Cairns & Lindinger, 2008; Cairns *et al.* 2015; Grassi *et al.* 2015). Furthermore, accumulation of metabolites and ions in muscle interstitium stimulates sensory feedback from group III/IV muscle afferents to the central nervous system, thereby inducing the sensation of muscle ache associated with intense exercise (Pollak *et al.* 2014) and providing a regulatory input for motor command centres (Amann *et al.* 2011). Maintenance of ion homeostasis is thus essential to sustain force production and power output during intense exercise.

Skeletal muscle has several systems that regulate ion homeostasis of which some are presented in Fig. 1. The figure also provides information about the effect of a period of SET on adaptations in ion transporters, exchangers, and channels as well as in metabolic enzymes of trained individuals. The importance of these adaptations for performance during intense exercise will be discussed in the following sections.

Effect of speed endurance training on muscle K^+ homeostasis

Intense exercise causes major perturbations in the rundown of transmembrane ion gradients that may reduce sarcolemmal excitability and impair force production of skeletal muscle (Cairns & Lindinger, 2008; McKenna *et al.* 2008). Specifically, contraction-induced accumulation of extracellular K^+ depolarizes sarcolemma, which causes inexcitability and muscle force decline in isolated muscle preparations (Clausen, 2013, 2015; Cairns *et al.* 2015). During intense exercise in humans, the extracellular concentrations of K^+ rise from 4 mM to critical levels (> 10 mM), which, based on studies in muscle preparations, contribute to development of muscle fatigue (Cairns *et al.* 1995, 1997; Cairns & Lindinger, 2008; McKenna *et al.* 2008). Accordingly, concentrations of K^+ have been shown to exceed 12 mM in muscle interstitial fluid during intense one-legged knee-extensor exercise and cycling when measured with the microdialysis technique (Juel *et al.* 2000; Nielsen *et al.* 2003, 2004*a*; Nordsborg *et al.* 2003; Gunnarsson *et al.* 2013). Interstitial concentrations of K^+ are likely to be much higher than that observed because of the low time resolution, geometric placement of probes (e.g. outside the transverse-tubular system), and probe-to-probe variation associated with the microdialysis technique (Green *et al.* 2000; Juel *et al.* 2000; Nielsen *et al.* 2004*a*). In support of this hypothesis, extracellular concentrations of K^+ as high as 50–80 mM have been reported in rat extensor digitorum longus muscle following repeated stimulations, which were associated with pronounced muscle force decline (Clausen, 2008, 2011, 2013). K^+ handling is therefore of importance for maintenance of sarcolemmal excitability during muscle contractions and any system that counteracts accumulation of extracellular K^+ could potentially delay fatigue development and enhance performance during intense exercise.

The Na^+–K^+ pump counteracts rundown of transmembrane gradients of Na^+ and K^+, and is essential for muscle functioning during sustained contractions (Clausen, 2015). Regulation of Na^+–K^+ pump activity is multifactorial, with activity being stimulated by muscle contractions, intracellular Na^+ concentrations, and hormones, such as adrenaline and insulin (Clausen & Flatman, 1977; Clausen & Kohn, 1977; Flatman & Clausen, 1979; Clausen, 2003). Even a few days of training are capable of increasing the abundance of muscle Na^+–K^+ pumps by ∼15% in recreationally active individuals (Green *et al.* 1993), and trained aged individuals have been shown to have a higher Na^+–K^+ pump abundance than untrained elderly subjects (Klitgaard & Clausen, 1989). Intensified training has been shown to increase the abundance of muscle Na^+–K^+ pumps by ∼10–70% (McKenna *et al.* 1993; Nielsen *et al.* 2004*a*; Bangsbo *et al.* 2009) when determined by both the [^{3}H]-ouabain-binding technique (Madsen *et al.* 1994; Evertsen *et al.* 1997) and Western blotting (Iaia *et al.* 2008; Bangsbo *et al.* 2009; Thomassen *et al.* 2010; Vorup *et al.* 2016). Even in already well-trained individuals, a period of SET effectively increases expression of the catalytic α subunits and structural β subunits of the Na^+–K^+ pump. For instance, in trained runners, SET, with a concomitant reduction in training volume, has been shown to elevate expression of Na^+–K^+ pump subunits α_1 (29%), α_2 (15–68%) and β_1 (10–15%) (Iaia *et al.* 2008; Bangsbo *et al.* 2009). The increased abundance of muscle Na^+–K^+ pumps induced by SET is associated with a lower rate of

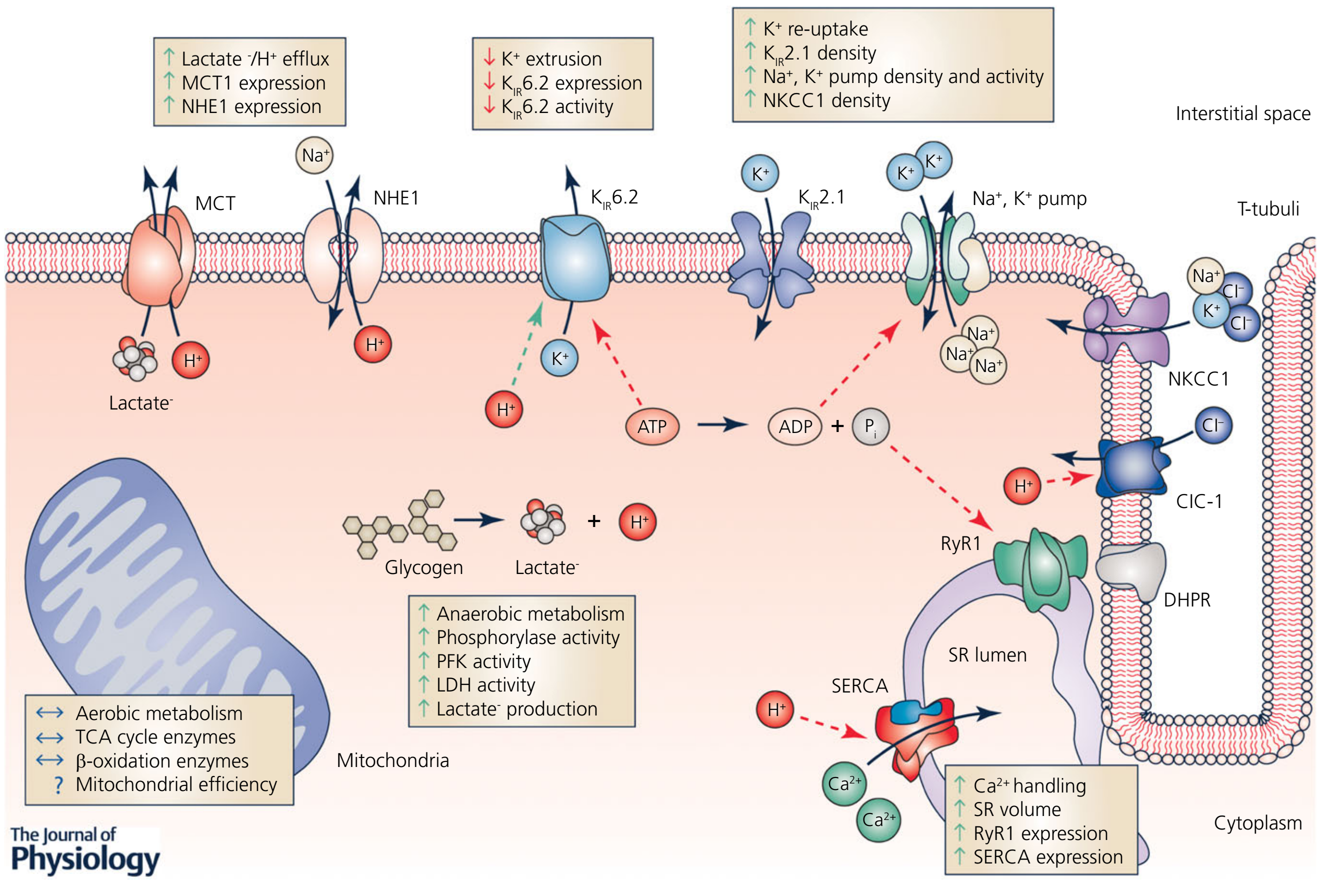

Figure 1. Overview of metabolic and ion handling systems in skeletal muscle of potential importance for performance during intense exercise, and their adaptations to a period of speed endurance training (SET)

Blue (continuous) arrows, direction of flux. Green (dashed) arrows, potentiation of a given protein/enzyme. Red (dashed) arrows, inhibition of a given protein/enzyme. Green (continuous) arrows, augmented by a period with SET. Red (continuous) arrows, lowered by a period with SET. Question marks (?), the effect of a period with SET is unknown. ClC-1, Cl^- channel isoform 1; DHPR, dihydropyridine receptor; $K_{IR}2.1$, K^+ inward rectifier channel 2.1; $K_{IR}6.2$, K^+ inward rectifier channel 6.2; LDH, lactate dehydrogenase; MCT, monocarboxylate transporter; NHE1, Na^+–H^+ exchanger isoform 1; NKCC1, Na^+–K^+–$2Cl^-$ exchanger isoform 1; PFK, phosphofructokinase; RyR1, ryanodine receptor 1, SERCA, sarcoplasmic reticulum Ca^{2+} ATPase; SR, sarcoplasmic reticulum; TCA, tricarboxylic acid cycle.

K^+ accumulation in arm venous blood during intense exercise and improved performance (Iaia *et al.* 2008; Bangsbo *et al.* 2009).

The question is, however, whether adaptations in muscle Na^+–K^+ pumps are essential for the improved performance observed following SET in trained individuals, since other studies did not find any change in Na^+–K^+ pump subunit abundances despite improved K^+ handling (Gunnarsson *et al.* 2013) and performance during intense exercise (Gunnarsson *et al.* 2012, 2013; Skovgaard *et al.* 2014). In trained cyclists, Gunnarsson *et al.* (2013) observed that 7 weeks of intensified training consisting of SET and aerobic high-intensity training, with a 70% reduction in training volume, increased performance by 18% during intense exercise (~4 min) and reduced femoral venous plasma K^+ concentrations by 0.7–0.8 mM in recovery from exercise, without alterations in expression of muscle Na^+–K^+ pump subunits. Thus, the lower femoral venous plasma K^+ concentration in recovery from intense exercise following the intervention period may be related to other factors than changes in muscle Na^+–K^+ pump content. In support of this proposition, maximal *in vitro* Na^+–K^+ pump activity of muscle homogenate has been shown to be augmented with no contaminant change in abundance of Na^+–K^+ pump subunits in endurance athletes after 3 weeks of aerobic high intensity training (Aughey *et al.* 2007).

One such factor that regulates Na^+–K^+ pump activity and is induced by SET is phospholemman (FXYD1), which interacts with the α and β subunits of the Na^+–K^+ pump (Crambert *et al.* 2002; Thomassen *et al.* 2010, 2016). Phosphorylation of FXYD1 increases affinity for Na^+ (lowers K_m) of the Na^+–K^+ pump and augments maximal ATPase activity ($\dot{V}_{max}$) (Crambert *et al.* 2002; Bibert *et al.* 2008; Rasmussen *et al.* 2008; Walas & Juel, 2012). In the aforementioned study in trained cyclists (Gunnarsson *et al.* 2013), expression of FXYD1 increased by 30% with the training period (Thomassen *et al.* 2016). In addition, non-specific FXYD1 phosphorylation was higher following the training at rest and during repeated intense exercise, mainly because of increased phosphorylation at serine68 (Thomassen *et al.* 2016), which regulates K_m of the Na^+–K^+ pump (Han *et al.* 2006; Juel *et al.* 2013). Similarly, 2 weeks of SET was shown to elevate phosphorylation of FXYD1 by 27% in elite football players, which correlated with enhancements in repeated sprint performance (10 × 20 m, 15 s active recovery) (Thomassen *et al.* 2010). Thus, SET-induced increases in expression and phosphorylation of FXYD1 may accentuate Na^+–K^+ pump activity and explain the lower femoral venous concentrations of K^+ in recovery from intense exercise observed by Gunnarsson *et al.* (2013).

Aside from Na^+–K^+ pumps, skeletal muscle expresses Na^+–K^+–$2Cl^-$ exchangers (NKCC1), strong inward rectifying K^+ channels ($K_{IR}2.1$), and ATP-sensitive K^+ (K_{ATP}) channels, all of which are involved in K^+ homeostasis of skeletal muscle and may be affected by SET (Spruce *et al.* 1985; Barrett-Jolley *et al.* 1999; Nielsen *et al.* 2003; Kristensen & Juel, 2010). However, only a few studies have investigated the effect of SET on muscle expression of NKCC1, $K_{IR}2.1$ and K_{ATP} channels in well-trained individuals. In trained runners, 4 weeks of SET was shown to increase expression of NKCC1 by 14% (Iaia *et al.* 2008). A higher expression of NKCC1 induced by SET may benefit ion homeostasis during intense exercise in two ways. Firstly, NKCC1 regulates myocellular volume via influx of K^+ and Na^+ with Cl^- and water (Gosmanov *et al.* 2003, 2004; Cairns *et al.* 2015), thereby counteracting the osmotic changes associated with ionic perturbations during exercise. Secondly, NKCC1-mediated influx of Na^+ stimulates Na^+–K^+ pump activity and thereby further re-uptake of K^+ (Clausen & Nielsen, 2007). In addition, 7 weeks of SET have been shown to increase expression of $K_{IR}2.1$ by 18% ($P = 0.06$) and to reduce expression of the K_{ATP} channel subunit $K_{IR}6.2$ by 14% in trained cyclists (Gunnarsson *et al.* 2013). A higher expression of $K_{IR}2.1$ may augment K^+ re-uptake when the electrochemical gradient for K^+ is reduced because of interstitial K^+ accumulation (Wallinga *et al.* 1999; Kristensen *et al.* 2006; Kristensen & Juel, 2010), whereas a lower density of K_{ATP} channel $K_{IR}6.2$ potentially reduces K^+ extrusion to the interstitium during intense exercise, since critically low myocellular ATP levels cause opening of $K_{IR}6.2$ (Davies, 1990; Davies *et al.* 1992, Xu *et al.* 2001). The concurrent increase in expression of $K_{IR}2.1$ and reduction in expression of $K_{IR}6.2$ may therefore also explain the lower femoral venous plasma K^+ observed in recovery from intense exercise and the improved performance observed by Gunnarsson *et al.* (2013). Nevertheless, the role of NKCC1, $K_{IR}2.1$ and K_{ATP} channels in K^+ handling and performance during intense exercise is not completely clear. Studies using muscle preparations indicate the importance of these channels, since pharmacological manipulation of NKCC1, $K_{IR}2.1$ and K_{ATP} channels affects fatigue development (Gong *et al.* 2003; Kristensen *et al.* 2006; Selvin & Renaud, 2015). And although muscle fatigue processes in non-functioning $K_{IR}6.2$ mice models are uncertain (Boudreault *et al.* 2010; Bruton, 2010), K_{ATP} channels are highly expressed in the transverse-tubular system (Nielsen *et al.* 2003) and have been shown to serve a myoprotective role against high cytosolic Ca^{2+} fluctuations and fibre damage during intense muscle activity (Selvin & Renaud, 2015).

As outlined above, SET elicits several adaptations related to K^+ handling of skeletal muscle in already trained individuals. SET may augment abundance and regulation of Na^+–K^+ pumps, increase expression of NKCC1 and $K_{IR}2.1$, and reduce expression of $K_{IR}6.2$. These adaptations are associated with lower concentrations of interstitial and femoral venous K^+ during exercise and a more rapid decline in recovery from exercise (Nielsen *et al.* 2004*a*; Gunnarson *et al.* 2013). Given that the concentrations of interstitial and femoral venous K^+ reach same levels at exhaustion after compared to before a period of SET, the adaptations related to K^+ handling induced by SET may, at least in part, explain the improved performance during intense exercise in already trained individuals.

Effect of speed endurance training on accumulation of lactate$^-$ and H^+

A major contributor to muscular energy production during intense exercise (½–6 min) is glycolysis, leading to production of lactate$^-$ and accumulation of H^+ (Hultman & Sahlin, 1980; Cheetham *et al.* 1986; Nevill *et al.* 1989; Gaitanos *et al.* 1993; Juel *et al.* 2004). Estimates of energy production derived from glycolysis range from 60–70% during 30 s of maximal cycling (Bogdanis *et al.* 1995; Kalsen *et al.* 2016) to 30–40% during 2–5 min of intense knee extensor exercise (Bangsbo *et al.* 1990, 1992). Thus, muscle lactate$^-$ may accumulate from ~1 to > 25 mmol (kg ww)$^{-1}$ and pH may decline from ~7.2 to < 6.6 during intense exhaustive exercise lasting a few minutes (Juel *et al.* 1990; Bangsbo *et al.* 1993; Hostrup *et al.* 2014). Skeletal muscle has several transport systems that facilitate efflux of lactate$^-$ and H^+ from myocellular compartments to the interstitium (Juel, 1998*a*; Juel & Halestrap, 1999; Pilegaard *et al.* 1999*b*). The main muscle transporters of lactate$^-$ and H^+ during exercise are the lactate$^-$–H^+ monocarboxylate cotransporters (MCT) of which the predominant isoforms 1 (MCT1) and 4 (MCT4) account for 70–80% of the lactate$^-$ and H^+ efflux (Mason & Thomas, 1988; Juel, 1997; Pilegaard *et al.* 1999*b*; Thomas *et al.* 2012).

While several studies have investigated the effect of SET on muscle lactate$^-$ and H^+ transport capacity and regulation in humans (Thomas *et al.* 2012), only few have been conducted in trained individuals. Unlike the marked improvements SET exerts in lactate$^-$–H^+ transport capacity of skeletal muscle in untrained and recreationally active individuals (Pilegaard *et al.* 1999*a*; Juel *et al.* 2004; Burgomaster *et al.* 2007), the adaptive response to SET in trained individuals is moderate and sometimes not evident. A few weeks of SET with reduced training volume have been shown to increase expression of MCT1 by ~10% in elite football players and by up to ~70% in trained men (Bickham *et al.* 2006; Thomassen *et al.* 2010; Gunnarsson *et al.* 2012; Puype *et al.* 2013), whereas no changes were observed in trained runners and cyclists (Bangsbo *et al.* 2009; Gunnarsson *et al.* 2013; Vorup *et al.* 2016). Unlike MCT1, expression of MCT4 does not seem to change with a period of SET in trained individuals (Bickham *et al.* 2006; Thomassen *et al.* 2010; Gunnarsson *et al.* 2013; Puybe *et al.* 2013; Skovgaard *et al.* 2014). However, it may be that alterations in MCT4 only occur in sarcolemma, since Dubouchaud *et al.* (2000) observed that endurance training increased expression of MCT4 in sarcolemmal fractions, but not in whole-muscle homogenate.

Apart from MCT, skeletal muscle has several systems that regulate myocellular H^+, including Na^+–H^+ exchangers (NHE1), carbonic anhydrase enzymes, HCO_3^- transporters, and H^+ buffers (Sahlin & Henriksson, 1984; Geers & Gros, 2000; Messonnier *et al.* 2007). While data on the effect of SET on carbonic anhydrase enzymes and HCO_3^- transporters are scant, studies have shown that SET increases content of NHE1 (Iaia *et al.* 2008; Skovgaard *et al.* 2014) and H^+ buffering capacity of skeletal muscle (Parkhouse & McKenzie, 1984; Bell & Wenger, 1988; Gibala *et al.* 2006; Baguet *et al.* 2011). NHE1 was shown to increase by 30–35% during a period of intensified training with reduced volume when SET was performed alone or in combination with resistance training in trained runners (Iaia *et al.* 2008; Skovgaard *et al.* 2014). Given that a higher expression of NHE1 may increase Na^+ uptake and H^+ efflux, it could be speculated that such adaptation may counteract contraction-induced accumulation of interstitial K^+ and sarcolemmal depolarization during intense exercise, since intracellular Na^+ stimulates Na^+–K^+ pump activity (Nielsen *et al.* 2004*b*) and H^+ accumulation causes opening of K_{ATP} channels and hence extrusion of K^+ (Davies, 1990; Davies *et al.* 1992, Xu *et al.* 2001; Street *et al.* 2005). While most studies found no change in *in vitro* muscle H^+ buffering capacity of muscle homogenate in trained individuals during a period of SET (Nevill *et al.* 1989; Iaia *et al.* 2008), *in vivo* muscle buffering capacity has been shown to be increased in some studies (Nevill *et al.* 1989; Gunnarsson *et al.* 2013). Thus, Nevill *et al.* (1989) observed no change in buffering capacity of muscle homogenate *in vitro*, whereas the estimated *in vivo* muscle buffering capacity from H^+ and lactate$^-$ during 30 s of maximal cycling was 44% higher after 8 weeks of SET in male and female runners. In line with this, Gunnarsson *et al.* (2013) observed that *in vivo* muscle buffering capacity estimated during intense exercise (~4 min) increased by ~18% with 8 weeks of combined SET and aerobic high-intensity training in trained cyclists. The difference between the *in vitro* and *in vivo* measurements observed may be due to a relative greater increase in the capacity of the trained muscle to release H^+ than lactate (Juel *et al.* 1998*a*,*b*), e.g. due to an increased expression of NHE1 (Gunnarsson *et al.* 2013), which *per se* will increase the estimated *in vivo* buffer capacity but not the *in vitro* equivalent.

The implications of adaptations in muscle lactate$^-$–H^+ transport and H^+ regulation for performance during intense exercise following a period of SET in trained individuals are not completely clear. While SET increases rate of H^+ release during intense exercise and lactate$^-$–H^+ clearance in the first few minutes of recovery from exercise, lactate$^-$ and H^+ formation greatly exceed muscle transport capacity during intense exercise (Bangsbo *et al.* 1990, 1993; Juel *et al.* 1990, 2004). To what extent accumulation of lactate$^-$ and H^+ contribute to fatigue development is also controversial (Korzeniewski & Zoladz, 2002; Pedersen *et al.* 2004; Bangsbo & Juel, 2006; Fitts, 2006; Lamb & Stephenson, 2006; Lindinger, 2007). Studies in muscle preparations have challenged the importance of lactate$^-$ and H^+ as muscle fatigue inducers (de Paoli *et al.* 2007), since low pH has little detrimental effect on fatigue development at physiological relevant temperatures, and lactate$^-$ has been shown to restore the depressing effect of raised extracellular K^+ on sarcolemmal excitability because of its inhibitory effect on Cl^- channels (ClC-1) (Nielsen *et al.* 2001; Pedersen *et al.* 2004; Hansen *et al.* 2005, 2005; de Paoli *et al.* 2010). In simulation models, Korzeniewski & Zoladz (2002) also demonstrated that H^+ accumulation may counteract disturbances in energy system integrity, especially oxidative phosphorylation in working muscles, in that glycolytic production of H^+ compensates the cytosolic alkalization caused by H^+ consumption of creatine kinase at the onset of exercise. Nevertheless, although these studies in muscle preparations *in vitro* and in simulation models imply that lactate$^-$ and H^+ may even act protectively against fatigue (Allen & Westerblad, 2004), studies in humans indicate that accumulation of lactate$^-$ and H^+ may be detrimental for performance during intense exercise. For instance, accumulation of lactate$^-$ induced by intense arm ergometer exercise was associated with greater accumulation of myocellular H^+ (pH: 6.65 *vs.* 6.82), reduced lactate$^-$ release (19 *vs.* 50 mmol), and a ~30% reduction in performance of the quadriceps during subsequent intense knee-extensor exercise (Bangsbo *et al.* 1996; Nordsborg *et al.* 2003). Notably, preceding arm ergometer exercise also caused a higher rate of extracellular K^+ accumulation during the subsequent knee-extensor exercise (Bangsbo *et al.* 1996; Nordsborg *et al.* 2003). The latter observation may be related to the greater myocellular accumulation H^+, thus causing opening of K_{ATP} channels (Davies, 1990; Davies *et al.* 1992, Xu *et al.* 2001; Street *et al.* 2005). In addition, lactate$^-$–H^+ transport and buffering capacity of skeletal muscle are good predictors for intense exercise performance (Nevill *et al.* 1989; Pilegaard *et al.* 1994; Bishop *et al.* 2003, 2004; Thomas *et al.* 2005; Messonnier *et al.* 2007), and athletes have higher lactate$^-$–H^+ transport and buffering capacity of skeletal muscle than trained and untrained individuals (McKenzie *et al.* 1982; Sahlin & Henriksson, 1984; Pilegaard *et al.* 1994; Edge *et al.* 2006). Lastly, β-alanine and sodium bicarbonate supplementation enhance intense exercise performance in trained individuals (Carr *et al.* 2011; Hobson *et al.* 2012). These effects are respectively related to increased muscle content of the H^+ buffer carnosine (Hill *et al.* 2007) and systemic HCO^{3-} concentrations (Sostaric *et al.* 2006). Augmented muscle carnosine content may also increase Ca^{2+} sensitivity and affect cross-bridge formation because of carnosine's potential role as a diffusible Ca^{2+}–H^+ exchanger (Dutka *et al.* 2012; Swietach *et al.* 2013), whereas increased systemic HCO^{3-} concentrations have been shown to reduce the rate of accumulation of extracellular H^+ and K^+ (Street *et al.* 2005; Sostaric *et al.* 2006). Collectively, these observations underpin the importance of improved muscle lactate$^-$–H^+ transport and H^+ regulation induced by a period of SET for performance during intense exercise in already trained individuals.

Effect of speed endurance training on sarcoplasmic reticulum Ca^{2+} handling and myofibrillar Ca^{2+} sensitivity

Skeletal muscle force production and relaxation are regulated by myoplasmic concentrations of Ca^{2+} and myofibrillar Ca^{2+} sensitivity. Force production is controlled by Ca^{2+} release via opening of sarcoplasmic reticulum (SR) ryanodine receptors (RyR1) and relaxation by subsequent re-sequestration of Ca^{2+} from the myoplasm into the SR, which predominantly is controlled by the Ca^{2+} uptake rate of SR Ca^{2+}-ATPase isoforms SERCAI and SERCAII, but also by the dissociation off-kinetics of troponin C, and the Ca^{2+} loading rate of Ca^{2+}-binding proteins (Westerblad & Allen, 1996*a*,*b*; Nogueira *et al.* 2013). Based on studies in muscle preparations, impairment of Ca^{2+} release in particular appears to contribute to muscle fatigue development (Allen *et al.* 2008, 2011; Nogueira *et al.* 2013). However, while the importance of Ca^{2+} handling for fatigue resistance is well described in muscle preparations, surprisingly few studies have investigated the effect of training on SR Ca^{2+} handling function and myofibrillar Ca^{2+} sensitivity in humans, and only two studies have, to our knowledge, been undertaken with SET in trained individuals (Ørtenblad *et al.* 2000; Thomassen *et al.* 2016).

Ørtenblad *et al.* (2000) observed that 5 weeks of SET augmented maximal $AgNO_3$ stimulated Ca^{2+} release in SR vesicles of muscle homogenate by 9%, which was associated with a 12% better mean power output during a repeated sprint test (10 × 8 s, 32 s recovery) in highly trained men. The adaptations in Ca^{2+} release function were presumably related to a training-induced increase in total SR-volume, since expression of RyR1, SERCAI and SERCAII was increased by SET, whereas $[H^3]$-ryanodine-specific binding remained unchanged (Ørtenblad *et al.* 2000). In spite of the increased expression of SERCAI and II, however, the authors observed no changes in SR vesicle Ca^{2+} uptake kinetics or Ca^{2+}-ATPase activity of muscle homogenate with the intervention. Because these measures were performed in *in vitro* preparations, extrapolation to *in vivo* Ca^{2+} release and uptake function during exercise is speculative. However, reduction in the *in vitro* SR Ca^{2+} release function of muscle homogenate has been shown to predict a decline in muscle torque after intense knee-extensor exercise of untrained individuals (Hill *et al.* 2001). In addition, 7 weeks of SET, combined with aerobic high-intensity training, was shown to increase total expression and level of phosphorylation of the SERCAII-regulatory protein phospholamban and Ca^{2+}–calmodulin-dependent protein kinase II (CaMKII) in cyclists (Thomassen *et al.* 2016). A higher expression and phosphorylation of phospholamban and CaMKII may increase the rate of SR Ca^{2+} release and re-uptake because of the reduced affinity for Ca^{2+} of SERCAII and increased opening probability of RyR1 (Suko *et al.* 1993; Rose *et al.* 2006; Kemi *et al.* 2007). Such effects might potentially improve SR Ca^{2+} handling during intense exercise. For myofibrillar Ca^{2+} sensitivity, currently no data are available regarding the effect of SET in trained individuals. In untrained individuals, Lynch *et al.* (1994) observed that sprint training lowered sensitivity for Sr^{2+}, while sensitivity for Ca^{2+} remained unaltered at normal pH but was lower at pH 6.6, which reflects the pH during intense exercise.

The limited number of studies that have investigated the effects of SET on Ca^{2+} handling makes interpretation of its implications difficult. Cross-sectional data have shown that endurance-trained individuals have lower *in vitro* Ca^{2+} handling function than resistance-trained and untrained individuals (Li *et al.* 2002). In line with this, aerobic training has been shown to reduce expression of SERCA and to lower *in vitro* Ca^{2+} release and uptake function of skeletal muscle (Majerczak *et al.* 2008; Green *et al.* 2011; Zoladz *et al.* 2013). Although these observations may imply that Ca^{2+} handling function is not the main determinant factor for endurance performance, studies have shown that Ca^{2+} uptake function may predict some of the variance during intense exercise (Harmer *et al.* 2014). Furthermore, 6 weeks of knee-extensor training was shown to increase rate of Ca^{2+} release and reduce Ca^{2+} leak from the SR, which were associated with enhanced peak power in untrained men (Munkvik *et al.* 2010). Nonetheless, while potential enhancing effects of SET in SR Ca^{2+} release function (Ørtenblad *et al.* 2000) as well as in expression and phosphorylation of phospholamban (Thomassen *et al.* 2016) may augment Ca^{2+} handling during intense exercise in trained individuals, more studies are needed on this issue.

Influence of speed endurance training on energy metabolism and reactive oxygen species: interaction with ion handling

Myocellular energy metabolism is important for excitation–contraction coupling during intense exercise. Na^+–K^+ pumps and SERCA account for ∼40% of energy expenditure of skeletal muscle during repeated contractions (Clausen, 2003; Walsh *et al.* 2006; Barclay *et al.* 2007) and any local limitation in energy availability and/or accumulation of metabolites impairs ion handling and accelerates fatigue development (Matar *et al.* 2000; Gong *et al.* 2003; Nogueira *et al.* 2013; Ørtenblad *et al.* 2013; Place *et al.* 2015; Selvin & Renaud, 2015). Low levels of ATP with a concomitant rise in free Mg^{2+}, H^+, ADP, IMP and P_i activates K_{ATP} channels (Allard *et al.* 1995; Barrett-Jolley & Davies, 1997; Pedersen *et al.* 2009), causes Ca^{2+}–P_i precipitation inside the SR (Allen *et al.* 2011), impairs Ca^{2+} handling (Li *et al.* 2002; Allen *et al.* 2008; Nogueira *et al.* 2013), and reduces Na^+–K^+ pump activity (Juel *et al.* 2014). Furthermore, anaerobic energy production through glycolysis leads to lactate$^-$ production and the accompanying H^+ accumulation changes SERCA activity and K_{ATP} channel sensitivity for ATP (Davies, 1990, 1992, Westerblad & Allen, 1993, Xu *et al.* 2001; Flagg *et al.* 2010; Nogueira *et al.* 2013). Anaerobic enzymes lactate dehydrogenase and creatine kinase may also interact with K_{ATP} channels (Crawford *et al.* 2002*a*,*b*), and Na^+–K^+ pump activity and SR Ca^{2+} handling are regulated by glycogenolysis and glycolysis (James *et al.* 1999; Okamoto *et al.* 2001; Dutka & Lamb, 2007; Nogueira *et al.* 2013; Ørtenblad *et al.* 2013). As such, a higher capacity for glycolytic ATP provision following a period of SET may potentially improve regulation of ion homeostasis during intense exercise.

SET has been shown to accentuate glycolytic enzymes (Dawson *et al.* 1998; Iaia *et al.* 2008; Puype *et al.* 2013: Vorup *et al.* 2016), increase glycogen content, and augment capacity for glycolytic energy production of skeletal muscle in trained individuals (Nevill *et al.* 1989; Shepley *et al.* 1992). Furthermore, given that CaMKII regulates the rate of glycogenolysis and glycolysis (Rose *et al.* 2006), a higher expression and phosphorylation of CaMKII induced by SET may accelerate the rate of glycolytic ATP provision during intense exercise (Gunnarsson *et al.* 2013; Thomassen *et al.* 2016). Notably, Iaia *et al.* (2008) observed that glycogen and phosphocreatine utilization during a second bout of intense exercise to exhaustion (∼1½ min) were lower after a period of SET, despite of higher accumulation of blood lactate$^-$ and longer time to exhaustion. This observation suggests that muscle oxidation and/or mechanical efficiency was improved by SET. Such adaptations may enhance metabolic stability and delay the impairments in contraction–excitation contraction coupling of skeletal muscle during intense exercise (MacIntosh & Shahi, 2011; Grassi *et al.* 2015). Furthermore, any potential glycogen-sparing effect of a period of SET might potentially counteract exercise-induced reductions in SR Ca^{2+} handling, since critically low glycogen levels in proximity to the SR have been shown to be associated with impaired Ca^{2+} handling function in highly trained individuals (Ørtenblad *et al.* 2011; Hostrup *et al.* 2014; Nielsen *et al.* 2014).

Intense exercise leads to formation of reactive oxygen species (ROS) and glutathione in skeletal muscle that reacts with thiol groups (glutathionylation) of various myocellular proteins of importance for ion handling (Juel *et al.* 2015; Cheng *et al.* 2016). Among these proteins, the Na^+–K^+ pump is subjected to glutathionylation in human skeletal muscle during intense exercise, which has been shown to attenuate maximal *in vitro* Na^+–K^+ pump activity by ∼25% in highly trained individuals (Hostrup *et al.* 2014; Juel *et al.* 2015); a phenomenon known as Na^+–K^+ pump inactivation (McKenna *et al.* 2008). Furthermore, ROS-production reduces SR Ca^{2+} release function and myofibrillar Ca^{2+} sensitivity (Place *et al.* 2015; Cheng *et al.* 2016). Interestingly, in recreationally active individuals, muscle Ca^{2+} release channel RyR1 is subjected to fragmentation following one session of SET, whereas endurance athletes have no fragmentation (Place *et al.* 2015). This observation indicates that training augments muscle oxidant defence and protects the integrity of proteins related to ion handling during intense exercise. In support of these observations, a few weeks of SET has been shown to increase activity of antioxidant enzymes glutathione peroxidase and glutathione reductase in recreationally active individuals (Hellsten *et al.* 1996; Bogdanis *et al.* 2013). In addition, Green *et al.* (1998) observed that 7 and 11 weeks of high-intensity resistance training attenuated reductions in Ca^{2+}-ATPase activity induced by intense cycling. While it could be speculated that SET would have similar adaptive effects in already trained individuals, future studies are warranted to elucidate whether SET augments muscle oxidant defence and counteracts glutathionylation of the Na^+–K^+ pump and fragmentation of RyR1 in the trained population.

Conclusion and perspectives

As presented in this review, only a few weeks of SET leads to significant performance enhancements during intense exercise in already trained individuals. Given that a period of SET has no apparent effects on $\dot{V}_{O_2,max}$, haemoglobin mass, plasma volume, muscle mass, capillarization or oxidative capacity in trained individuals, the enhancements in performance during intense exercise induced by SET may be related to muscle adaptations in ion handling, including improved regulation of K^+ homeostasis, enhanced lactate$^-$–H^+ transport, and in some instances augmented H^+ buffering capacity and SR Ca^{2+} handling. The implications of these adaptations with relevance for performance during intense exercise may involve:

- Lower rate of accumulation of interstitial K^+ of exercising muscles due to increased Na^+–K^+ pump activity and repression of K_{ATP} channels.
- Reduced extrusion of K^+ from K_{ATP} channels because of augmented lactate$^-$–H^+ transport and H^+ regulation.
- Augmented Ca^{2+} handling.
- Higher capacity for glycolytic ATP provision.

Because of the marked perturbations of multiple ions (Cairns & Lindinger, 2008; McKenna *et al.* 2008) and their interactions with energy production and ROS formation during muscle activity (MacIntosh & Shahi, 2011; Place *et al.* 2015; Cheng *et al.* 2016), fatigue processes during intense exercise in trained individuals are complex, and further studies are needed to elucidate the exact mechanisms underlying the enhancing effects of SET on performance. For instance, recent observations in muscle preparations indicate that extracellular Ca^{2+} interacts with lowered transsarcolemmal K^+ gradients during muscle contractions, where a reduction in extracellular Ca^{2+} lowers the intracellular activity of K^+ and contributes to fatigue development (Cairns *et al.* 2015). In this perspective, studies are warranted that investigate the importance of muscle Ca^{2+}–H^+ and Na^+–Ca^{2+} exchangers, as well as ClC-1 channels for exercise performance in humans and their potential adaptability to training. For example, ClC-1 channels are of interest because of their role in regulating Cl^- conductance and sarcolemmal excitability during muscle activity (Dutka *et al.* 2008; de Paoli *et al.* 2010, 2013). Furthermore, improvements related to ion handling, lactate$^-$ transport, and H^+ regulation after a period of SET could possibly affect regulation of myocellular and plasma osmolarity during exercise, and the implication of such changes for performance should be studied (Lindinger *et al.* 1995, 2013). Despite the difficulty in recruiting athletes for advanced physiological training interventions, more studies are needed in the athlete population. Lastly, it is relevant to probe the long-term effects of SET, since the intervention period of most studies that have provided an emphasis for this training modality only lasted a few weeks.

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Additional information

Competing interests

None declared.